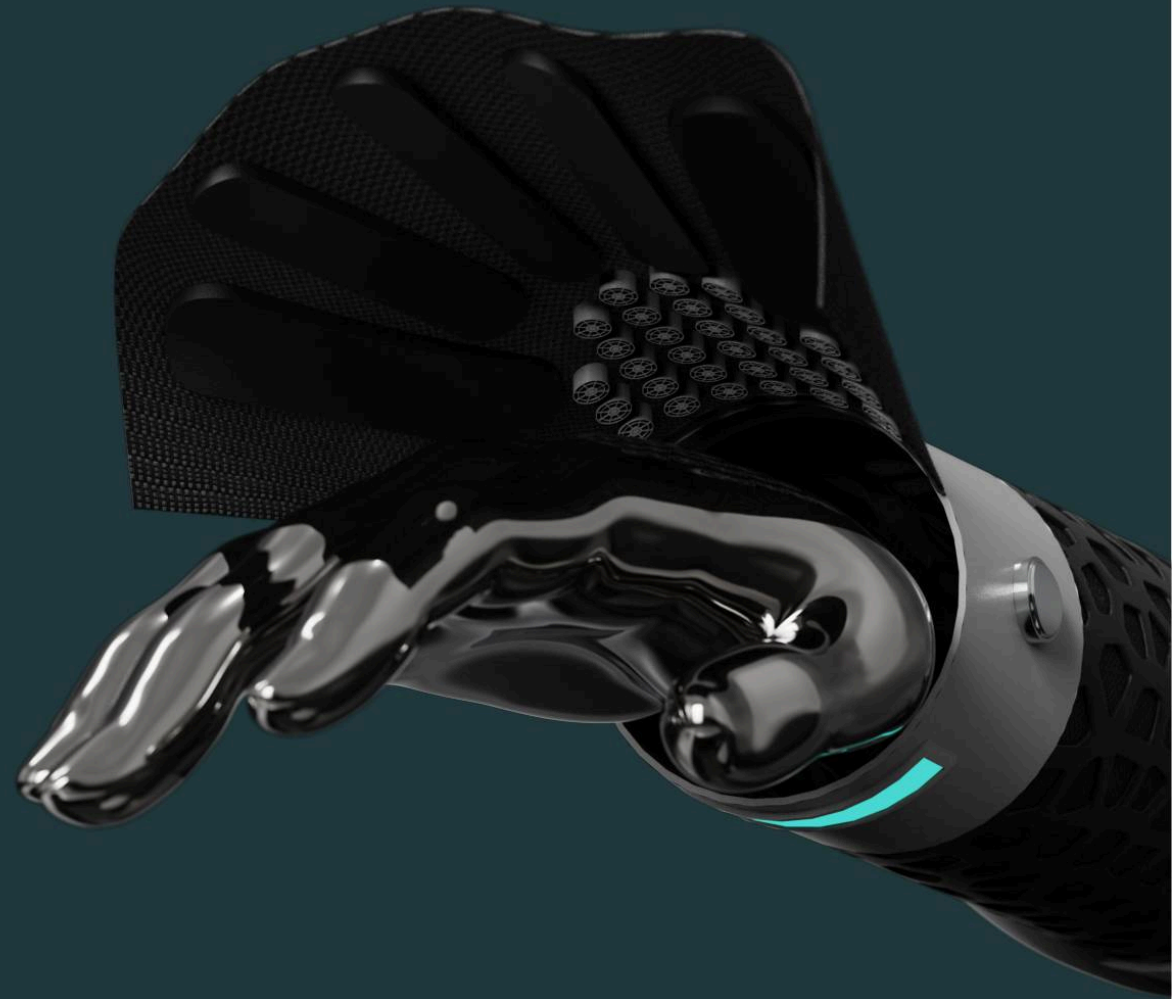




Interax.



Focus • Future • Innovation

Introduction

Abstract

Growing rates of diagnoses and prescription requests indicate a profound need for management strategies for ADHD in all areas of life, but particularly in the classroom. Fidgeting allows the myriad distractions on an ADHD person's mind to fade into the background, facilitating deeper and extended focus. Harnessing this, Interax aims to provide ADHD students a deeper understanding of their focus. This is enabled through the developing technologies of biometric data collection and processing, electro muscle stimulation, and ultrasonic skin haptics, which in 30 years time will be ready to be combined into one sleek package providing instant and tailored haptic responses to simulate fidgeting. This combined with presenting the user with easy to understand metrics about their focus and reflection questions allows students to develop meaningful strategies to improve their focus.

Meet the Team



**Sinai
Rhodes**
02274683



**Freddie
Nicholson**
02092811



**Emmanuel
Irechukwu**
02271458



**Liberty
Wright**
02033078



**Kaitai
Yang**
02218028

Acknowledgements

Thank you to our tutor Freddie Page for his consistently insightful advice, and our module leads Maria Apud Bell and Sheraz Arif for their feedback. Thank you to Dr Hamid Farhani, Dr Yunlong Zhao, Katarina Poole, and Angus Nicholson for taking the time to talk to us, helping Interax become what it is.

Contents

3	Contextual Trends
4	Forecasting ADHD
5	Teaching for ADHD
6	User Needs
7	Design Requirements
8	Project Development Plan
9	Haptic Feedback
10	Biometrics
11	Focus ML Algorithm
12	User Reflection
13	Concept Overview
14	Concept Development
15	System Overview
16	Power, Safety & Durability
17	User Interface
18	Final Concept
19	Economics
20	Evaluation
21	Project Management
22	References

Contextual Trends

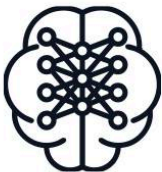
Trends in ADHD Treatment



Diagnoses and prescriptions for ADHD medication have increased over the past two decades. [1]



Interventions have been introduced to help reduce disruptive behaviour and it has been found self regulating methods are most effective. [2]



New treatment options such as Neurofeedback have been developed but it's effectiveness is not yet proven. [3]

Current Limiting Factors

Recruitment and retention challenges have impacted teachers' ability to provide support for ADHD students. [4]



Strain on budget has limited schools in supporting ADHD students effectively. Recent 1bn funding boost impact remains to be seen. [5]

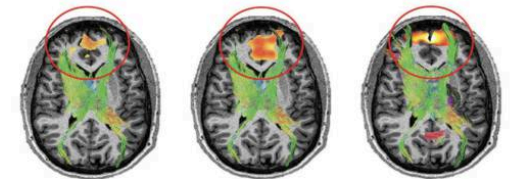


Technology enablers have only recently emerged, and the development of SEND products requires multiple approvals throughout the process.



Focus and ADHD

Research from the Association for Behaviour Analysis International[6] found that fidget devices improved on-task behaviour in the classroom, with immediate performance enhancements observed upon use.



ADHD ADHD + Fidget Device Neurotypical

Researchers from the University of Auckland and Mātai Medical Research Institute found MRI scans showed that fidget devices could improve activation of ADHD student's decision-making.

Current Issues with Fidget Devices

Distraction to Others Learning

Fidget devices have previously been criticised for distracting other students, leading some schools to ban them from classrooms entirely[8]. While they were originally marketed as focus aids, concerns over their disruptive potential have sparked debate about their effectiveness.

Lack of Evidence

There is limited research on fidget devices improving classroom focus and learning outcomes. While some studies suggest benefits, findings remain inconclusive. With advancements in machine learning and biometrics, there is greater potential to develop a research-backed solution.

Lack of Feedback

Current fidget devices are low-cost 'toys' that are mass manufactured and do not have any capability of giving user feedback. There has been a rise in smart tech such as fitness trackers and a similar model could be applied to fidget devices.

Forecasting ADHD

ADHD is a neurological condition characterised by patterns of inattention, hyperactivity, and impulsivity that are more severe, frequent, or interfering than typically observed in individuals at a comparable level of development. A key aspect of ADHD involves neurobiological differences, including altered levels of neurotransmitters, which are chemicals in the brain that facilitate communication between neurons.

Serotonin Levels in ADHD: ADHD students typically have lower serotonin levels compared to their neurotypical peers. Serotonin is crucial for mood regulation, decision-making, and neurological function. In the context of ADHD, reduced serotonin levels can contribute to difficulties in managing impulses and emotions, affecting the overall cognitive function and behaviour. This imbalance can lead to various challenges[1].

Current Methods to Help ADHD



Reliance on Medication

Medication, predominantly stimulants like methylphenidate and amphetamines, remains a cornerstone of ADHD management in educational settings[2].

While medication can help manage symptoms, it does not address underlying cognitive or behavioural issues associated with ADHD and medications do not work uniformly across all individuals.



Inadequate Behavioural Intervention

Behavioural interventions are recommended alongside medication, but the implementation and quality can be inconsistent across schools[3].

Many schools lack the resources to provide comprehensive behavioural support and the effectiveness of behavioural strategies can vary widely depending on the school's commitment.



Insufficient Learning Adjustments

Although educational adjustments and accommodations are crucial for students with ADHD, there is often a lack of truly personalised learning approaches[4].

Educational adjustments such as extra time are helpful but do not fully accommodate the diverse needs of ADHD students. Technology has the potential to offer more personalised learning experiences.

Futures Scenario

By 2055, NHS advancements will enable earlier and more accurate ADHD diagnoses. Alongside medication, enhanced diagnostic tools and digital health technologies will promote self-regulation as a key component of treatment, empowering ADHD patients to actively manage their symptoms more effectively.

Project Definition

WHO

ADHD students from the age of 14 and above. This is because we believe that it is important for the students themselves to be able to reflect.

WHAT

Implementation of integrated educational technologies and non-pharmacological interventions designed to support ADHD students.

WHERE

Initially targeted for implementation within UK educational institutions, due to the centralised healthcare system (NHS), however potential for global adoption.

WHY

These methods aim to provide ADHD students with personalised learning experiences that address their unique needs it is adapted to the specific student.

WHEN

We anticipate our product will be fully operational by 2054, providing essential support to help students with ADHD enhance their focus and engagement in educational settings.

HOW

Employing Electrical Muscle Stimulation (EMS) and ultrasound skin haptics to simulate the tactile sensation of objects, combined with machine learning algorithms that analyse biometric data to monitor focus.

Teaching for ADHD

By looking at how teachers are trained to create a welcoming environment and improve educational outcomes, we can consider how to assist students in developing learning strategies to equip them for work and lifelong learning.

Across academic and educational documents, three key areas were identified for fostering an environment where those with ADHD can focus and thrive

Support

ADHD students often find a lot of rules not designed for them, and can therefore become defensive when teachers tell them off. The ADHD Foundation advise teachers to practise positive reinforcement, encouraging “good behaviour” and ignoring “bad behaviour”.

Criticism of the student, no matter how constructive, can trigger Rejection Sensitive Dysphoria (RSD) causing significant emotion dysregulation, and leading to a vicious cycle of “poor behaviour” and attempted discipline[1].

Distractions

Teachers are encouraged to sit students away from doors and windows, and in between calm high achieving students.

They are also advised to break down tasks into small chunks to make them more accessible to ADHD students, both in initially approaching them and also so if they do lose focus then they can jump back in without having missed too much[2].

Movement

Current advice to teachers suggests “giving students a valid reason to move around” eg. if there’s a classroom task to do that will get them out of their seat the teacher should give that task to an ADHD student.

Stress balls are also considered as an option for helping a student manage their own fidgeting, they are specifically encouraged as they are relatively silent and don’t have too many distracting functions[2]

Interview with a Teacher

We interviewed Angus Nicholson, who has been teaching mathematics and design technology for over 20 years. The full interview transcript can be seen in our meeting minutes folder. Mr Nicholson’s comments aligned heavily with our research into the three categories above.

“we try and break down the tasks, so keeping people focus is good and we also have lots of movement around

these little gadgets or fiddle toys are a good thing, but it equally it would be quite useful to have an ability to mute them

We try to provide instant feedback. Longer term feedback, but making sure it's direct to the children

We don't sort of draw attention to what the child's doing”

User Needs

This user needs analysis highlights key challenges faced by students with ADHD and suggests practical, discreet tools and strategies to improve focus, organisation, and classroom engagement.

Organisation

- **Issue:** Students with ADHD often struggle to maintain structure in their daily routines, leading to poor organisation[1].

- **Solution:** Implement consistent schedules and introduce simple, portable tools that can serve as reminders or cues to help students stay on track[1].

Transitions

- **Issue:** Transitions between activities or changes in routine can cause anxiety and disrupt focus for students with ADHD[2].

- **Solution:** Provide advance notice of schedule changes and incorporate engaging, familiar objects to ease the shift between activities, promoting smoother transitions[2].

Fidgeting

- **Issue:** Physical restlessness and fidgeting can hinder concentration and disrupt classroom activities[3].

- **Solution:** Encourage the use of discreet, tactile items that allow for controlled physical activity, helping students stay grounded while maintaining focus[3].

Focus

- **Issue:** Students with ADHD may experience fluctuating moods and difficulty maintaining attention during lessons[1].

- **Solution:** Introduce calming techniques and focus aids that provide subtle sensory feedback, helping regulate mood and sustain attention throughout the lesson[1].

Rewards

- **Issue:** Maintaining motivation can be challenging for students with ADHD, leading to inconsistent performance[3].

- **Solution:** Use personalised reward systems that integrate interactive elements, reinforcing positive behaviours through, hands-on tools[3].

User Journey



Ben (18)

Student with ADHD



"I really struggle to focus in class, but I want to improve. At the moment I don't feel like I control my focus, it controls me."

Design Requirements

Based on our user interviews, we identified three key pain points that impact students with ADHD in the classroom. From these insights, we formulated HMW (How Might We) questions to guide our design thinking

	Maintaining Focus in Class	Managing Fidgeting	Reducing Anxiety
Key Problems	Students with ADHD experience fluctuating moods and difficulty maintaining attention during lessons, making it challenging to stay engaged. <i>"Sometimes I try really hard to focus, but my mind just drifts off without me even noticing."</i> <i>"I start a lesson feeling fine, but after a while, I just can't sit still and listen anymore."</i>	Uncontrolled fidgeting can hinder concentration and disrupt classroom activities, leading to a loss of engagement. <i>"I know I need to stop tapping my desk, but if I don't move, I just can't think properly."</i> <i>"I wish I had something small to fidget with that doesn't get me in trouble."</i>	Transitions between activities can cause anxiety and disrupt focus, making it hard for students to smoothly adjust to new tasks. <i>"When the teacher suddenly says, 'Okay, time to move on,' I feel stuck and don't know what to do next."</i> <i>"I hate when we switch activities too fast—I need time to get ready for the next thing."</i>
Opportunities	How might we use haptic feedback to provide subtle, real-time sensory cues that help students sustain focus without being distracting?	How might we design a wearable that allows for controlled, non-disruptive fidgeting while reinforcing focus through tactile feedback?	How might we create a haptic-assisted transition system that provides calming and structured guidance during shifts between activities?

3 Key Design Requirements

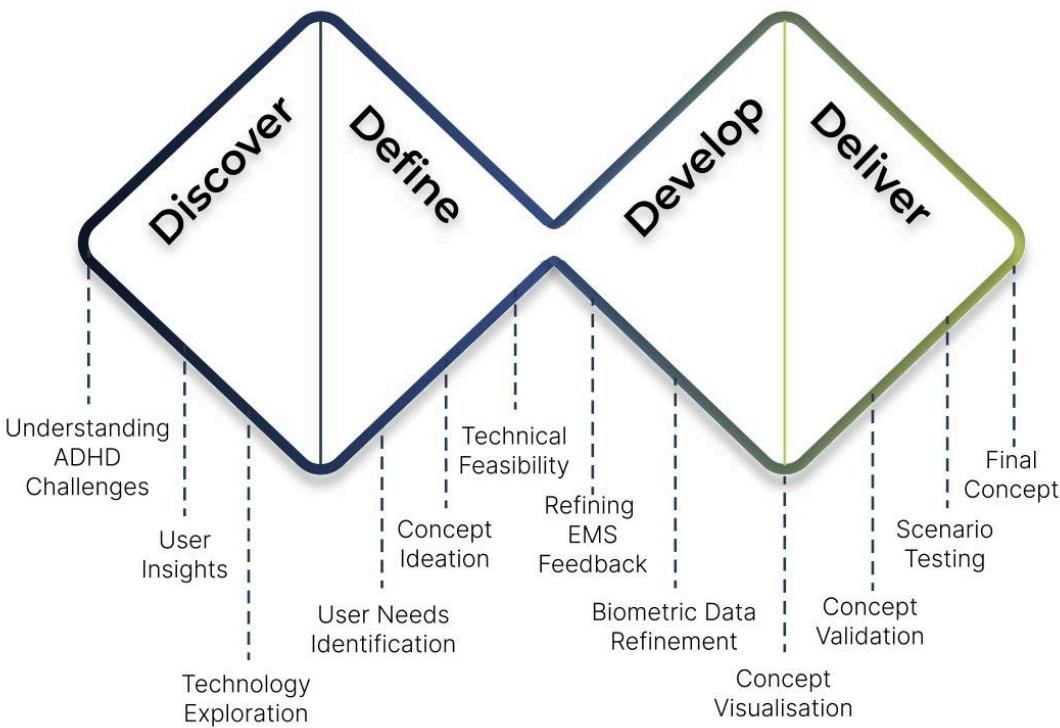
Non-disruptive	Adaptive	Sustain Focus
 ADHD students already battle cognitive overload; a subtle tool keeps them engaged without adding distractions.	 The system should learn from the user's focus patterns, biometric data and fidget preferences to personalise responses	 Feelings of being overwhelmed leads to disengagement; so a real-time support should help keep students on track.

Project Development Plan

This project develops a biometric-enabled fidget bracelet using electrical muscle stimulation (EMS) to help ADHD students manage focus. Through user research, technology validation, and iterative design, we refined the concept to be discreet, effective, and tailored to student needs. This page outlines our development process, key insights, and next steps in bringing this solution to life.

Positioning and Next Steps

The Double Diamond framework structures our design process through four phases: Discover, Define, Develop, and Deliver, ensuring thorough problem-solving and innovation. By iterating through research, refinement, and validation, we develop a biometric-enabled fidget bracelet tailored to ADHD students, enhancing focus and engagement.



Term 1 Development

Building on feedback from Term 1, we refined our approach by narrowing our focus to a more streamlined, effective solution. Initially, our concept explored multiple integrated ideas, but we recognised the need for a more targeted design.

Through iterative development, we identified key elements that best supported our goal: creating a discreet, personalised focus aid. This refinement process enhanced functionality, ensuring our solution was both effective and adaptable to user needs. By prioritising user experience and feasibility, we developed a more cohesive, validated system, setting the foundation for implementation and future development.

Engineering Design Process

Area	Process	Tools	Reasoning
Technology Assessment	Evaluate the feasibility of EMS feedback and biometric tracking for focus measurement. Identify technological limitations and refine sensor integration.	Research papers, expert consultations, technology benchmarking.	Ensures EMS and biometric tracking are effective, safe, and suitable for ADHD students.
Concept Development	Generate concept sketches, explore form factors, and refine the bracelet's interaction model. Ensure discreet, comfortable, and engaging design.	Figma for UI/UX, Blender for 3D modelling, CAD software, user sketches.	Helps in refining a product that balances functionality, comfort, and usability.
Biometric Data Refinement	Analyse physiological responses to EMS and refine feedback loops to maintain engagement without overstimulation.	Data modelling, Python for biometric data analysis, UX research.	Enhances personalisation and effectiveness of the focus-tracking mechanism.
Concept Validation	Conduct feedback sessions with ADHD students to test comfort, EMS sensation, and perceived effectiveness in maintaining focus.	Interviews, surveys, prototype testing, user feedback.	Ensures the product is intuitive and beneficial without being distracting.

Haptic Feedback

Recent research indicates that hand fidgeting can enhance concentration by helping manage restlessness. However, spontaneous hand movements may distract visual attention, which is especially problematic in academic settings. To address this challenge, we propose using haptic feedback to facilitate controlled hand fidgeting for ADHD students. This approach aims to harness the concentration benefits of fidgeting while minimising visual distractions, thereby creating a more supportive learning environment.

Electrical Muscle Stimulation

Electrical Muscle Stimulation (EMS)

is a technique that uses controlled electrical impulses to trigger muscle contractions by mimicking the natural signals sent from the brain.



EMS-based haptic rendering delivers tactile feedback by electrically stimulating muscles to contract, simulating the reaction force

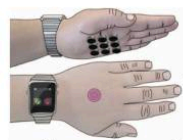
one feels when touching a real object. EMS contracts specific muscles to counteract the user's motion. For example, when a user presses a virtual object, stimulating the forearm's extensor muscles prevents the hand from moving through the object, thereby creating a realistic haptic sensation.



In our project, EMS is used to simulate the gripping function of the hand by providing resistance feedback to the muscles. However, EMS cannot deliver the precise tactile perception required at the fingertips, so we need to introduce new technologies.

Ultrasound Haptic Technology

Ultrasound haptics is a non-contact technology that generates tactile sensations by focusing ultrasound waves onto the skin. An array of ultrasound transducers emits high-frequency sound waves, which can be precisely manipulated to create focal points.



By dynamically adjusting the position and intensity of these focal points, the system produces controlled pressure patterns on the skin that simulate the feeling of touch or interaction with virtual objects. This innovative approach enables users to experience detailed tactile feedback without the need to wear any devices, making it highly suitable for applications in virtual and augmented reality, interactive displays, and remote communication interfaces.



We introduce ultrasound haptics technology to deliver precise tactile feedback that complements EMS. This integration of multimodal haptic simulation offers users an immersive fidgeting experience.

Electrical Muscle Stimulation

Improved resolution and fidelity. Sub-10 ms response is achievable, force-feedback illusions refined and included in high-end setups.

Ultrasound Haptic Technology

Improved Spatial Resolution: Smaller, more efficient ultrasound transducer arrays enable sharper mid-air tactile feedback.

By the late 2030s, the number of discrete stimulation points (electrodes) will likely reach hundreds or thousands, enabling near-realistic tactile holograms and very fine force control.

By 2045–2055, we anticipate resolution reaching the level of human tactile acuity. This means haptic systems might recreate the feel of any material or object at the fingertips.

High-Density Arrays: Significantly denser transducer layouts produce complex, realistic tactile patterns.

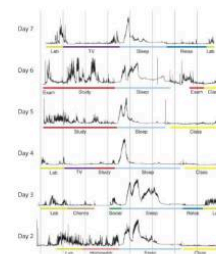
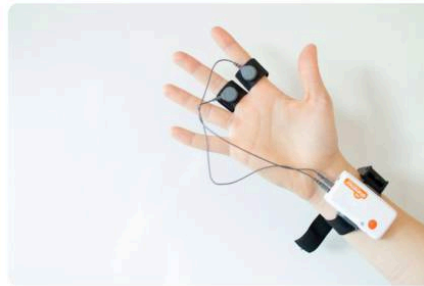
Ultra-Realistic Haptics: Tactile experiences rival physical touch, allowing fully immersive holographic interactions.

Biometrics

Skin Conductivity / Electrodermal Activity (EDA)

What is Electrodermal Activity?

Electrodermal activity (EDA) describes the property of the human body that causes variations in skin characteristics, primarily due to sweat gland activity. GSR (Galvanic Skin Response) is the specific sensor used to measure these variations, detecting changes in skin resistance that correspond to physiological or emotional states[1].



How does EDA relate to focus?

The paper Analysis of Electrodermal Activity Signal Collected during Visual Attention Oriented Tasks[2] found that EDA linked to the human visual attentiveness in the classroom. The research found that EDA outperformed other similar methods tested by other researchers. The research found that EDA offers a non-invasive way of obtaining real-time data of student focus.

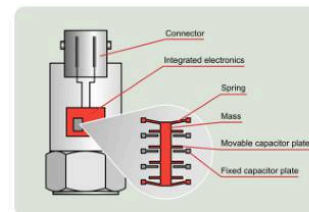
Respiratory Rate

How does it relate to focus?

A slower breathing rate enhances focus by regulating the noradrenaline system, which supports attention and learning. Controlled breathing releases noradrenaline in balanced amounts, improving mental clarity, cognitive function, and emotional stability. This helps students think more clearly, stay engaged, and process information more effectively in class[3].

Sensors Used

A Piezoelectric or MEMS-based respiratory sensor will monitor breathing rate by detecting subtle skin expansions and contractions on the forearm. By measuring pressure or strain changes caused by respiratory-induced vascular fluctuations. This provides a reliable and non-intrusive way to track breathing without relying on sensitive motion detection.



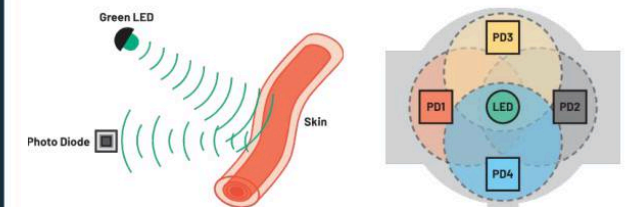
Heart Rate

How does it relate to focus?

Increased heart rate has been linked to heightened brain alertness during learning[4]. Research suggests that as individuals engage in tasks requiring focus, their heart rate spikes, indicating a direct connection between cardiovascular activity and cognitive engagement.

Sensors Used

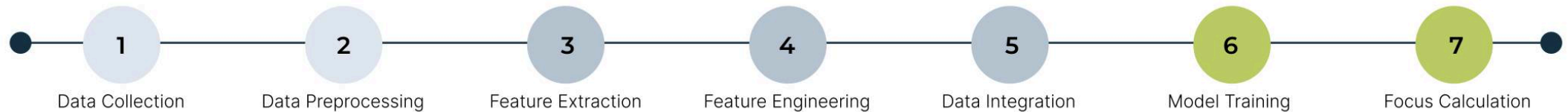
Photoplethysmography (PPG) sensors measure heart rate and blood volume changes by detecting variations in light absorption through the skin. Typically using infrared or green light, they enable non-invasive, continuous monitoring in wearables. Multiple photodiodes (PDs) around a central LED enhance accuracy by capturing light from different angles, reducing noise and power consumption while improving measurement consistency[5].



EDA, heart rate, and breathing rate together provide input data for our machine learning system. Experimenting with control and test groups allows us to fine-tune the model for accurate focus measurement.

Focus ML Algorithm

Although the focus metric has been previously discussed, the detailed process behind its calculation remains unexplored. This section provides a step-by-step breakdown of how the machine learning algorithm computes this focus metric from the biometric data.



Data Handling

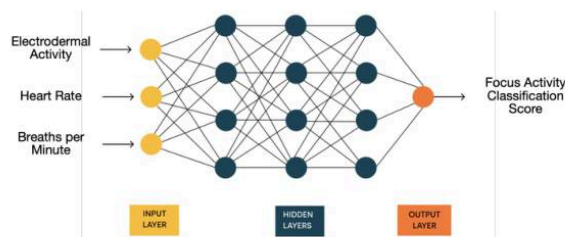
- 1 Biometric data is collected from the Interax sleeve using a PPG sensor, a GSR sensor, and a MEMS-based respiratory sensor. The acquired data undergoes noise filtering to ensure only relevant signals are processed. Additionally, normalisation procedures standardise the range of various biometric signals, facilitating their comparison and integration on a uniform scale. This prepares the data for subsequent feature selection, engineering, and integration processes.
- 2

Feature Refinement

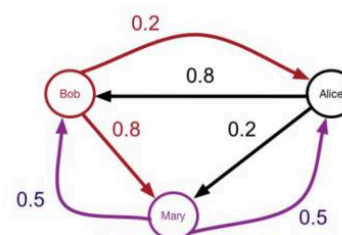
- 3 From the PPG sensor, fluctuations in heart rate variability are extracted, while the GSR sensor provides data on changes in EDA amplitude, and alterations in breathing patterns are captured by the respiratory sensor. During feature engineering, these inputs are combined to create composite features that reflect the interplay between physiological signals, enhancing the model's ability to detect nuanced changes in focus. In the data integration phase, these engineered features are consolidated into a cohesive dataset, which is then standardised to ensure consistency across different types of biometric data, preparing it for effective analysis and model training.
- 4
- 5

Finalised Model

- 6 During the model training phase, a neural network is utilised to analyse the integrated dataset, making it particularly suited for identifying patterns from the synthesised biometric features. This model is complemented by the use of Markov chains, which are instrumental in predicting transitions between states of focus based on probabilistic trends. The focus metric itself is computed from the neural network's output, quantifying the user's level of concentration as a numerical value. This output is continuously refined through model recalibration using incoming user data, which ensures that the metric remains accurate and adapts to changes in user behaviour and environmental factors. This dynamic adjustment helps maintain the precision and applicability of the focus assessment over time.
- 7



An initial neural network prototype can be developed using PyTorch and publicly available datasets which was measured whilst users were doing work related tasks.



Markov chains are a mathematical graph method used in machine learning to predict future states based on probabilities. This method could be effectively utilised in the pattern/trend analysis feature of our mobile application, allowing the use of pre-determined states of focus to estimate if the user is in the right mindset to learn.

User Reflection

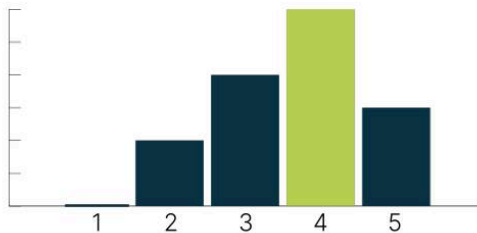
A key component of Interax is the use of reflection, both to train the algorithm which determines the focus metric, and as a tool in itself to help students understand their own focus.

When finishing a focus session the app will prompt the student to respond to a few questions, ranking and reflecting on their focus. This will help them recognise key distractions and develop tools to help.

LoFi Prototyping

A form was made and sent out to students and adults who work from home who were encouraged to fill it in repeatedly after periods of focus. It asked respondents to rate their focus, internal and external distractions, and reflect on how they might avoid them in the future. Respondents were then asked to what extent they felt reflecting on their focus, and the accountability the form provided helped in developing their focus.

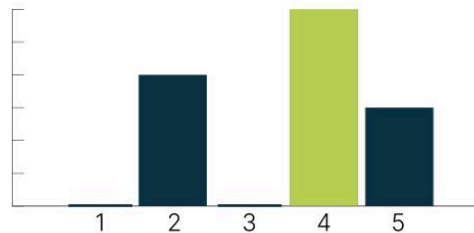
To what extent do you feel that reflecting on your focus has helped you understand how you could focus more in the future?



"When i'm leaving a lecture I don't have time to reflect, but when I do I've forgotten how I felt."

Respondents found reflecting on their focus helpful but struggled with remembering to fill in the form, or having enough time to do it properly. Users opposed to the concept of reflection found it less helpful.

To what extent do you feel that knowing you would fill in this form held you accountable to focus during the activity?



"A form won't hold me accountable because i'm the one filling it in. If someone else was rating me then maybe that would be better"

Respondents found the accountability aspect of the form less effective, especially the student respondents who as a demographic struggle more with accountability.

Case Study on Finch: Self Care Pet

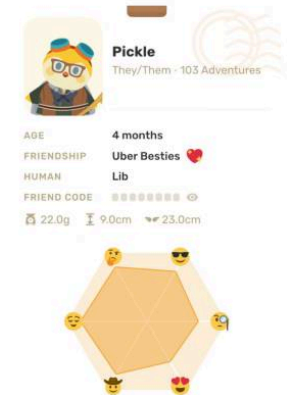


Finch uses gamification, self reflection, and accountability via a pet to encourage self care. It has an average of 4.9/5 stars from 39k reviews on the app store at time of writing. We are looking at the features and reviews of this app to develop how we can use self reflection to help our users build focus tools.

External Accountability:

Finch uses the growing personality of your pet to encourage users to complete self care activities through positive reinforcement and perceived support.

"Having a little friend helps so much. With Finch I always want to complete my goals"



Reflection:

"research suggests that enhancement of self reflection ability is the first critical process in patient recovery (from mental health issues)" [1]. Finch provides in depth and more accessible reflection prompts to help users understand they're feelings and create coping mechanisms. This approach to developing tools can also be applied to productivity and focus.

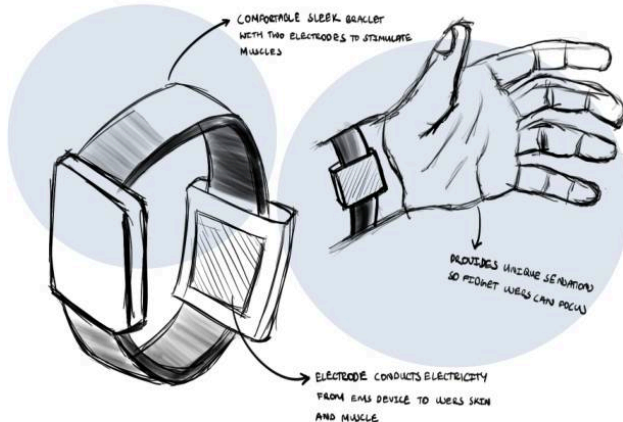
"I have ADHD and Finch has helped me to be so much more productive and organised."

Key Design Developments

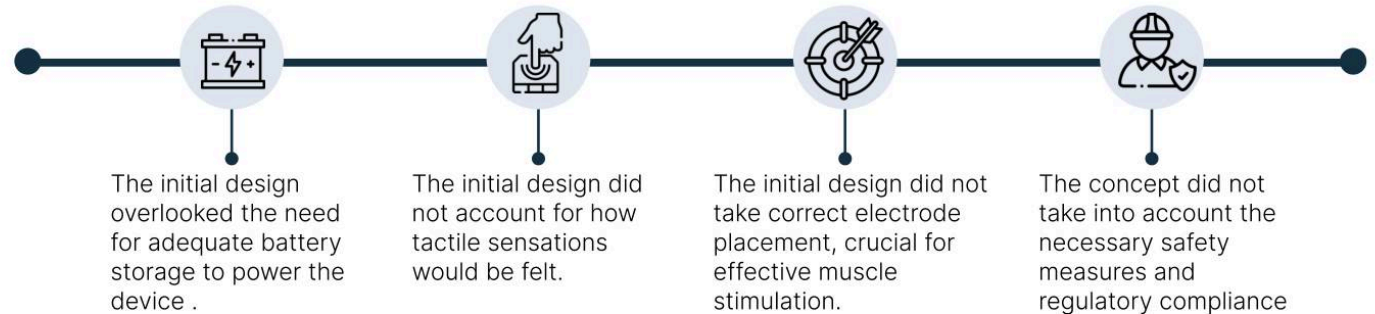
- Initial reflection should be very quick after the focus period to record a snapshot for later reflection.
- Deep reflection should be an opt in component of the experience, as it is most effective for those who complete it in good faith.

Concept Overview

The initial concept stemming from first term involves a wearable bracelet equipped with two electrodes that make contact with the skin. This bracelet can electrically stimulate muscles to generate various tactile sensations, mimicking the experience of using a physical fidget device.



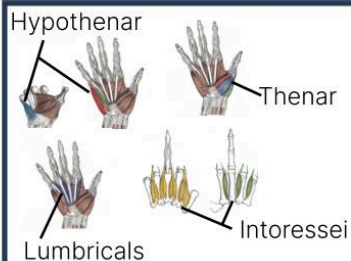
Key Issues with Initial Concept



Concept Development

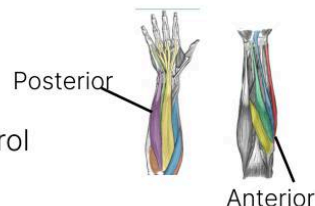
Increased understanding of EMS technology revealed that the precise placement of electrodes is crucial to ensure effective stimulation of the target muscles [1].

Muscles Needed to Control Hand



Intrinsic muscles – located within the hand itself. They are responsible for the fine motor functions of the hand. It consists of 4 main muscles the thenar, hypothenar, lumbricals and intoressei [2].

Extrinsic muscles – located in the anterior and posterior compartments of the forearm. They control crude movements and produce a forceful grip [2].



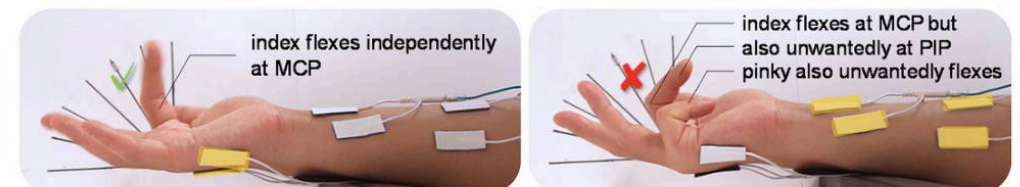
From the research done where these muscles are located there should be electrodes placed on each muscle, however further validation was carried out ensuring that this was the case.

Concept Validation

Through literature review an article was found titled “Increasing Electrical Muscle Stimulation’s Dexterity by means of Back of the Hand Actuation”[3].

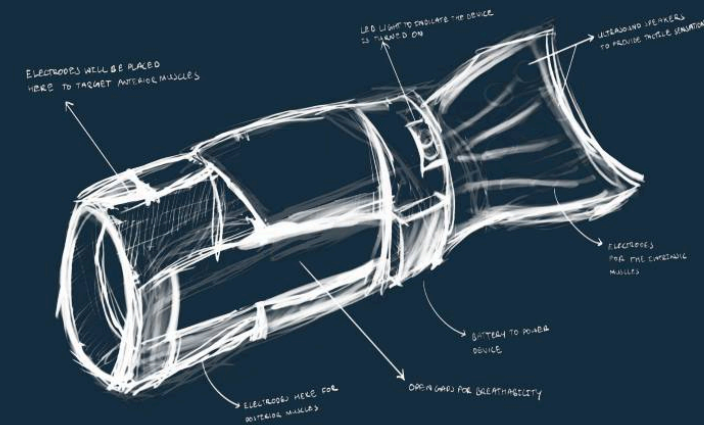
The article looked at refining the dexterity of EMS by introducing a new electrode placement on the back of the hand. This novel layout targets the interossei and lumbrical muscles in the palm, previously overlooked for EMS applications

This approach proved more effective than traditional EMS setups, which often result in simultaneous and undesired flexion of multiple fingers due to less precise electrode placement on the forearm. By focusing on the back of the hand, the research promises more refined and independent control of finger movements, suitable for advanced haptic and interactive systems



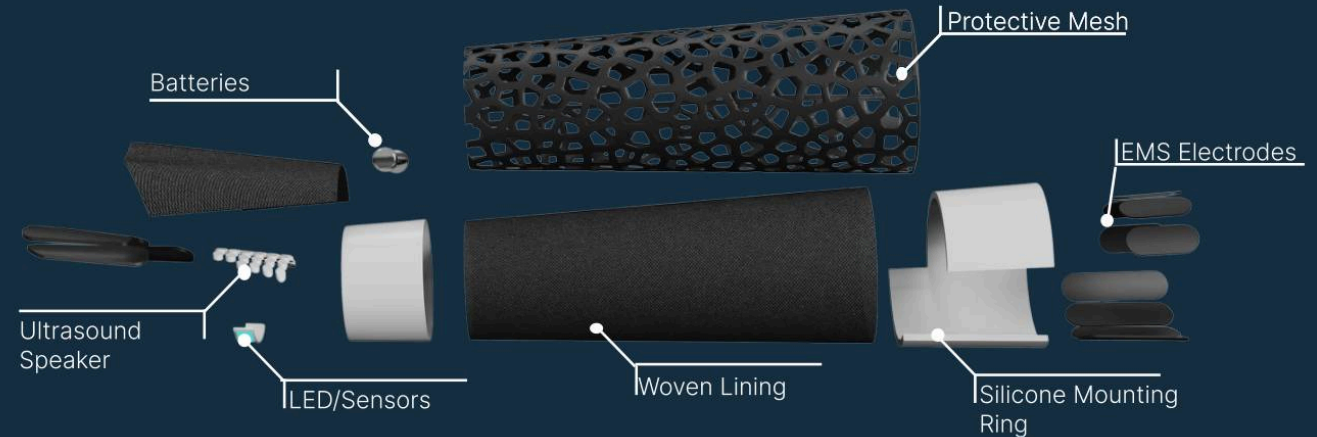
Concept Development

Using insights from validation gained from literature review, a developed concept sketch was done. Experts and target users were spoken to which then led to, a detailed 3D model being produced.



The Interax sleeve is a sleek, discreet wearable designed to seamlessly integrate into daily life, even under clothing. Engineered for comfort and adaptability, it utilises EMS and ultrasound feedback to enhance focus through subtle, non-intrusive cues. Biometric sensors track key physiological metrics for personalised, real-time engagement.

Product Teardown



Ultrasound Speaker: will be composed of piezoelectric ceramics the speaker rims will be coated in medical-grade silicone to prevent skin irritation and enhance comfort during prolonged wear.

EMS Electrodes: a hydrogel-coated conductive fabric will be used. Hydrogel will be used due to its excellent conductivity, flexibility and ability to maintain adhesion over repeated use. It is easy to remove and a simple saline solution can be used to refresh the adhesive.

LED: a standard surface-mount LED housed in a polycarbonate for soft illumination. The LED indicates power status and mode changes

Woven Lining: a breathable conductive textile, such as silver-coated nylon for flexibility and durability it provides a comfortable layer between the user's skin and the electrodes while also aiding in electrical conductivity and sweat-wicking

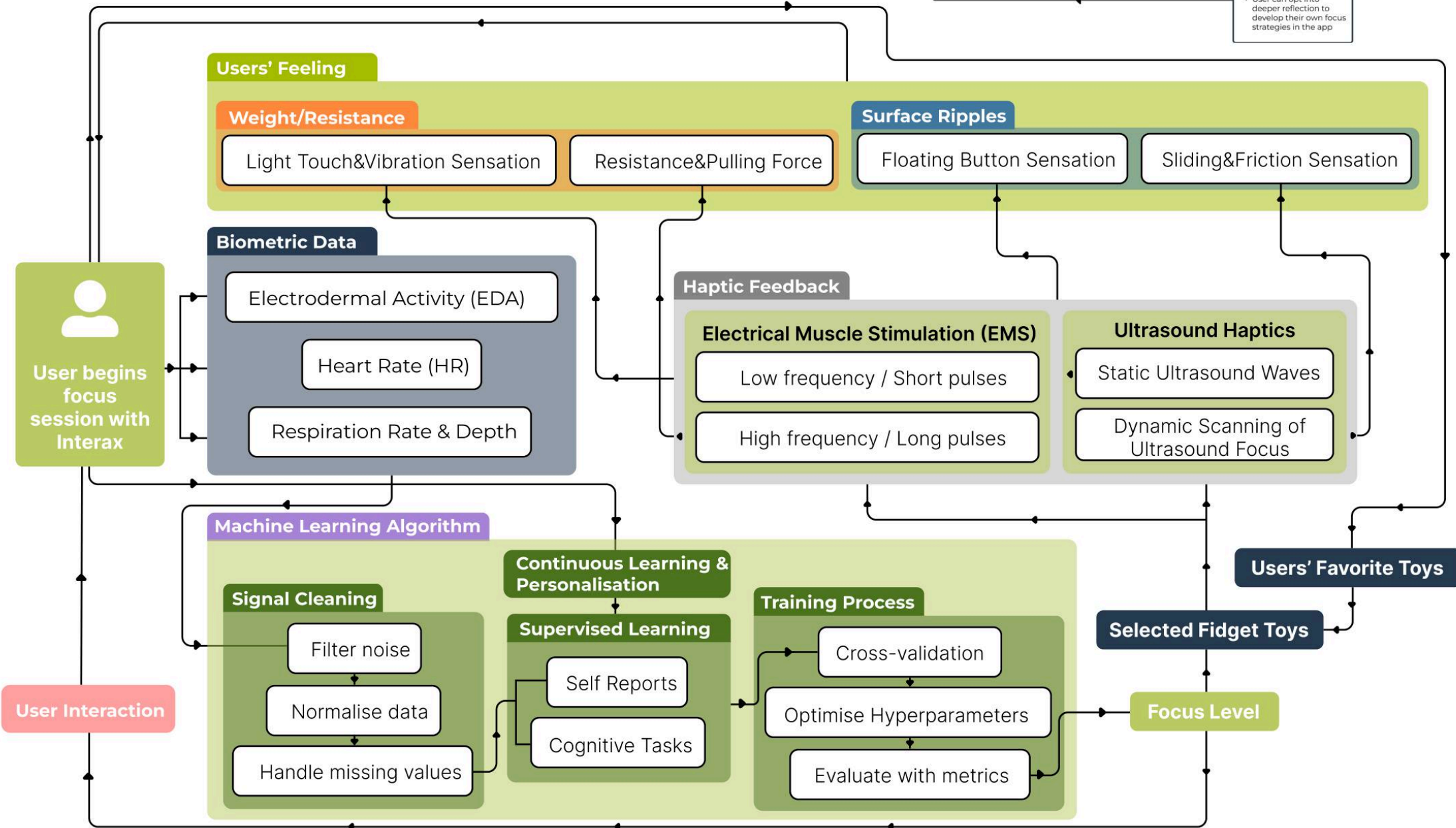
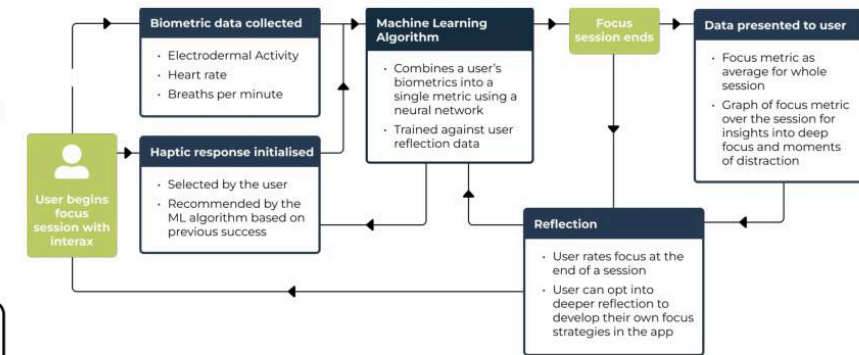
Protective Mesh: 3D-printed TPU for a durable yet lightweight structure it acts as a protective exoskeleton for internal components while maintaining airflow and flexibility

Mounting Rings: Medical-grade silicone is used to secure the device comfortably to the user's arm

System Overview

Taking our design requirements and insights from research into our tech enablers, a system diagram was developed.

Initial Diagram:



Power, Safety & Durability

Battery

Interax uses a modular battery-swapping system, allowing users to swap out depleted batteries for fully charged ones, eliminating downtime. This approach is inspired by professional camera batteries and e-bike power systems, ensuring continuous use without waiting for recharging.

Power Draw

Based on the EMS2000 Manual[1], EMS electrodes operate at **9V**, with each drawing **20mA** (0.02A). The system consists of 14 electrodes and using the equation below the resultant resulting EMS power consumption is **2.52W**.

$$Power(W) = Voltage(V) \times Current(I)$$

According to an MIT study on ultrasonic haptic feedback[2], each transducer operates at **5V** and draws **10mA** (0.01A). The system consists of **30 transducers** and using this same equation we can calculate approximately **1.5W** draw.

In addition to EMS and ultrasound, biometric sensors and wireless communication contribute to power consumption. Biometric tracking (including heart rate and EDA monitoring) consumes **0.3W**, while Bluetooth adds **0.2W** bringing the total power draw to 4.52W.

To ensure practical wearability, battery size was selected based on average 14-year old wrist circumference data of approximately 152mm[3], **30mm-wide batteries** fit comfortably on either side of the wrist. The chosen battery model, Lier -6530271, has a voltage of **3.8V** and a capacity of **430mAh** (0.430Ah)[4]. Providing a combine **5.96Wh** Interax would run for approximately **79.2 minutes** on one charge.

30-Year Projection

According to Statista's projections[5], global battery storage capacity for electricity generation is expected to increase from 70GW in 2022 to 945GW in 2050, representing a **13.5x increase**. Given this, the estimated runtime with projected 2050 battery capacity is **17.82 hours** allowing Interax to comfortably exceed the required operational time for a full school day.

Safety

Interax leverages EMS technology, which is already widely used in FDA-approved over-the-counter muscle stimulators and rehabilitation devices. These devices typically operate at currents of **10–100mA**, which is well within the safe range for human use according to clinical studies.

- The maximum current used in Interax is **20mA**, aligning with commercial EMS devices designed for consumer use [1].
- The voltage range is **5–50V**, ensuring effective stimulation without exceeding safety thresholds[1].
- EMS operates at frequencies between **20–100Hz**, which falls within the safe neuromuscular stimulation range used in medical and sports applications[6].

Additionally, Interax incorporates automated intensity regulation, ensuring that stimulation levels remain within safe and comfortable thresholds based on user feedback and data. This prevents over-stimulation or discomfort, making the device suitable for long-term use.

Durability

The interax is designed with modularity in mind, allowing users to easily disassemble components for cleaning and replacement.



The woven lining can be removed and washed separately, ensuring hygiene without compromising durability.

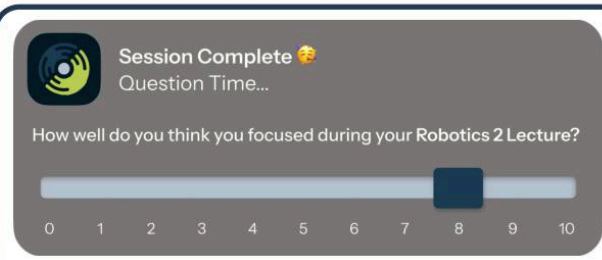
The removable battery system prevents long-term degradation of internal electronics. By using swappable components, users can maintain their device without requiring frequent full replacements making the Interax sleeve a more sustainable and cost-effective solution.

User Interface

Interax is designed to seamlessly integrate into the student's routine, providing a structured yet flexible way to improve focus. The app connects with the Interax sleeve, allowing students to personalise their focus sessions, receive real-time feedback, and track their progress. By collecting biometric data and user input, the app refines its recommendations over time, adapting to individual needs. The UI is designed to be intuitive, with a clean and simple layout that prioritises ease of use. Users can set up their focus session, select response types, and reflect on their engagement. The dashboard provides insights into focus patterns, helping students build long-term strategies for maintaining attention.



The app provides real-time biometric tracking, displaying metrics such as focus level, heart rate, and respiratory rate, to help users understand how different conditions impact their engagement. Over time, patterns emerge, allowing users to make informed adjustments to their focus strategies.

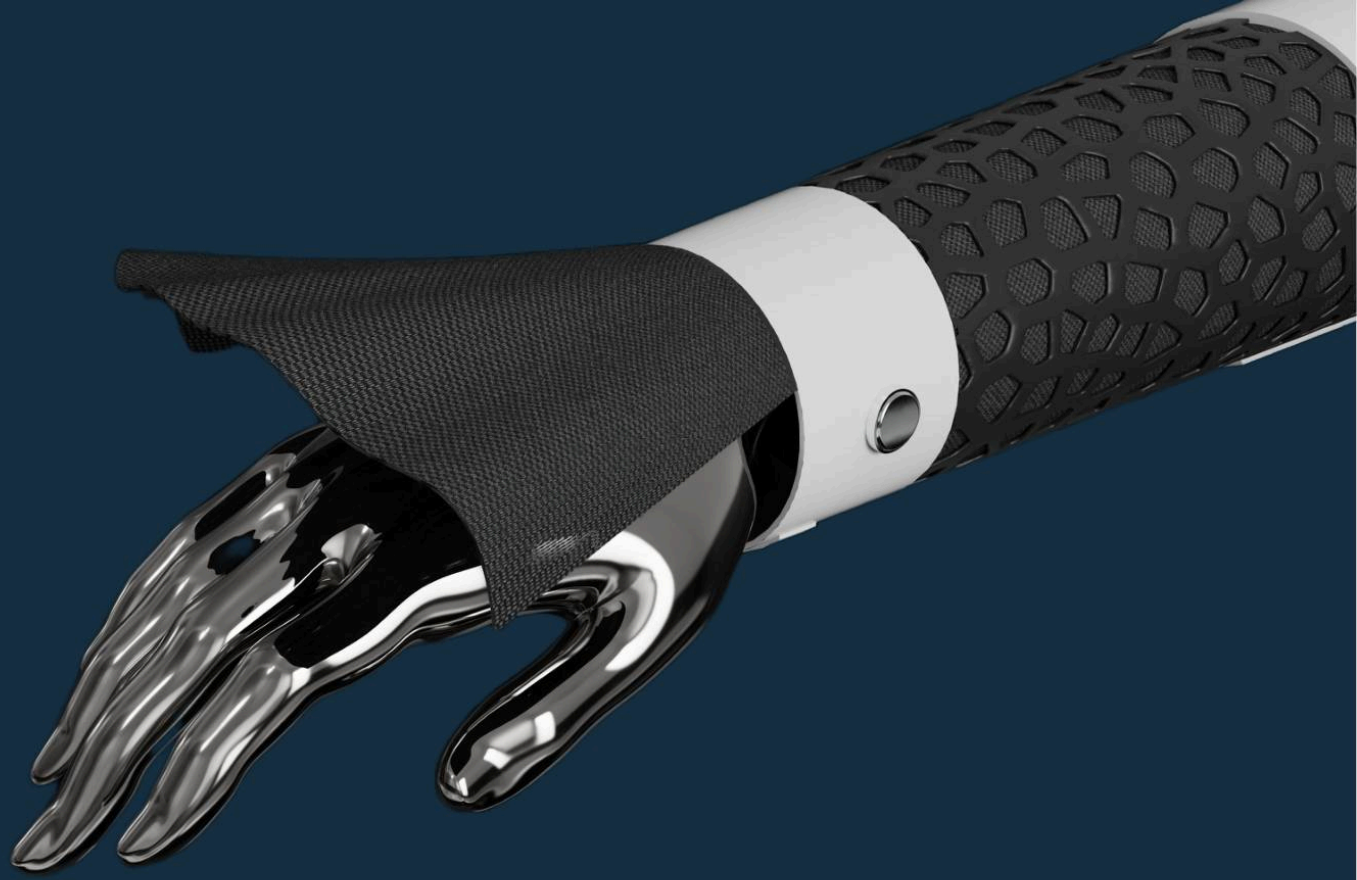
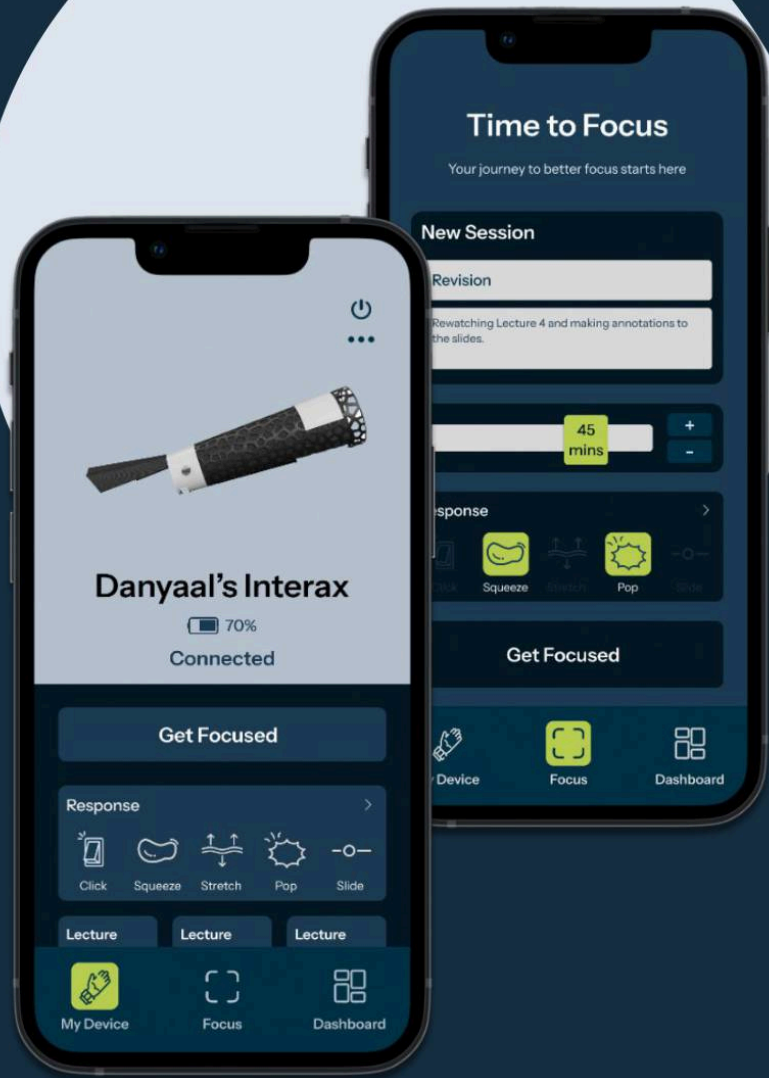


The reflection widget prompts users to evaluate their focus after each session. This feature is based on research showing that structured self-reflection enhances focus awareness and helps individuals develop strategies for improvement.



The response widget mimics common fidget behaviours like squeezing and popping to keep users engaged. The machine learning algorithm tailors response types and resistance levels based on user preferences and biometric data for a personalised experience.

Final Concept



Interax.

Economics

Market Analysis

The Disabled and Elderly Assistive Technology market has been expanding with a CAGR of **6.7%** [1] between 2022 to 2030. Approximately **2.6M** students in the UK are diagnosed with ADHD with **27%** [2] of those being children under the age of 18. This shows a clear market share for which we can target our product in it's initial stages. After our discussion with Dr Hamid Farhani, we also discovered that more children are being diagnosed after testing due to Social Media awareness. UK Government has also invested within the SEND space with a **£740mn** [3] investment in 2024 towards schools to help students with special needs integrate in the classroom.

Unique Selling Points (USPs)

- 1



Personalised fidget response allows the user to tune the device to fit their requirements.
- 2



Combined Electrical Muscle Stimulation and Ultrasound to create a sensation not felt before.
- 3

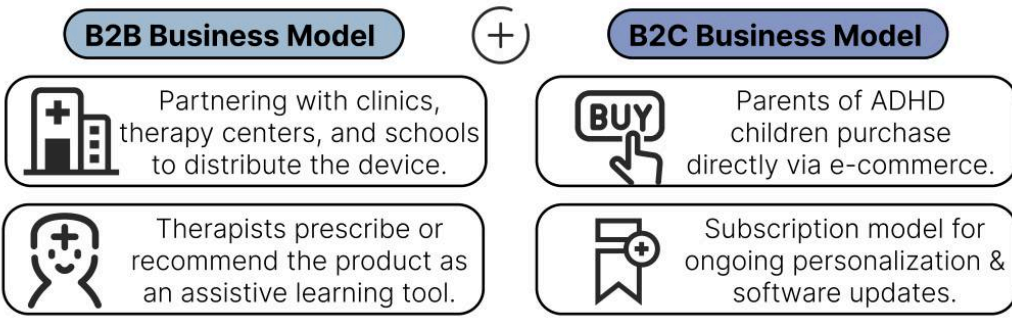


Neural network used to adapt to the user's previous data to accurately classify their focus levels on the fly.
- 4



Privacy conscious federated learning ensures that no user's individual data can be identified.

Business Model



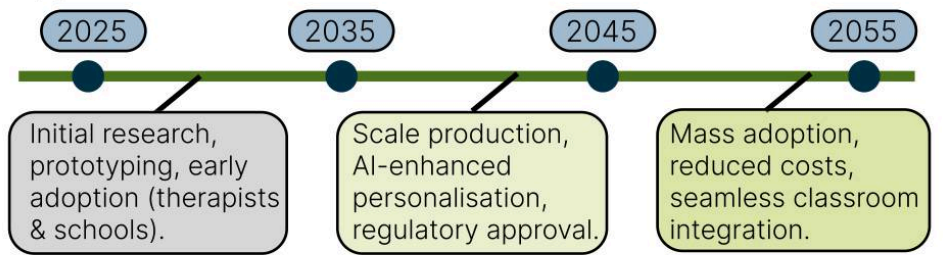
Projected Cost Reduction

	2025	2055
Conductive Fabric Base	£15	£2
EMS Electrode Pads & Circuitry	£150	£40
Electrodermal Activity Sensors	£120	£65
Ultrasound Haptic Transducers	£100	£50
Battery System and Power Management	£10	<£5
Assembly and Labor	£100	£2
	£495	£164

By 2055, mass production reduces the unit cost from £495 to £164, enabling consumer (B2C) and institutional (B2B) pricing strategies that sustain profitability while broadening market access and overall affordability. Pricing our product at **£249** will give a profit of approximately **£85**.

Feasibility Evaluation

Our solution targets full-scale deployment by **2055**, integrating EMS, ultrasound haptics, and ML. We ensure cost reduction through B2B&B2C partnerships, alternative funding, and scalable production, making the device **affordable, adaptive, and future-proof for ADHD support**.



Evaluation

Expert Validation

Dr Hamid Farhani - ADHD Specialist



Hamid noted that social media has raised ADHD awareness, encouraging more individuals to seek diagnoses. He suggested incorporating reminders and feedback into the app while emphasising the importance of more precise prescriptions tailored to genetic factors.

Angus Nicholson - Teacher for 20+ years



Angus highlighted the importance of leveraging digital tools to enhance accessibility and engagement, particularly for younger audiences. He also suggested incorporating personalised feedback mechanisms while emphasising the need for clear data to aid decision-making.

Conclusion

Through extensive research, we identified a growing need for non-medicinal ADHD management. Using iterative design, desk research, and evaluation, we developed Interax: a non-invasive, adaptive, and effective solution that enhances classroom engagement for students with ADHD.

Risk Mitigation

Criteria	Risk	Chance	Impact	Mitigation
Sensor accuracy affected by motion artefacts	Medium	High	Inaccurate readings may affect focus tracking reliability.	Use advanced filtering algorithms to compensate for motion artefacts.
Discomfort from prolonged wear	Medium	High	Users may experience discomfort, discouraging consistent use.	Select hypoallergenic materials and ensure ergonomic design.
Data privacy concerns with biometric monitoring	High	High	Users may feel uneasy about data security and sharing.	Implement strong encryption and allow user control over data access
Potential delays in real-time data processing	Medium	Medium	Delays in feedback could reduce effectiveness.	Optimise processing algorithms for faster response times
Variability in individual responses affecting accuracy	High	High	Different physiological responses may lead to inconsistent data.	Personalise baseline calibration to account for user variability

1. Non-Disruptive

Interax ensures a seamless classroom experience with invisible muscle stimulation and haptic feedback, minimising distractions. The slimline sleeve, worn discreetly under clothing, helps ADHD students focus without being singled out.

2. Adaptive

Interax's machine learning algorithm adapts to each student's focus patterns, refining support strategies over time. This ensures personalised and effective concentration across various classroom settings.

3. Sustains Focus

By combining instant fidget-based responses with long-term cognitive strategies, Interax provides both immediate relief and lasting improvements. This empowers students to manage their focus independently.

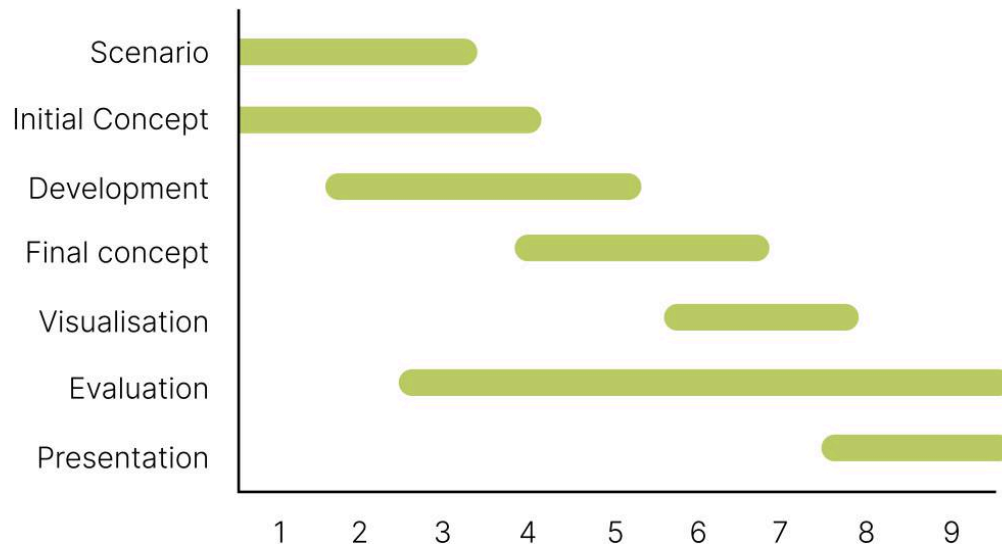
Project Management

Team Reflection

Reviewing our project management from last term and with deeper understanding of how we work as individuals and as a team, we began this term with a clean slate. Regular meetings, both with and without Dr Freddie Page, aided communication, whilst clear meeting minutes held us all to account.

Overall, we felt that the project gained much more clarity and focus following the interim report. What began as a somewhat unclear narrative spanning multiple market segments has been refined into a specialised device designed to help ADHD students focus in the classroom.

Gantt Chart



Collaboration Tools

We used Figma for real-time brainstorming and design, with Google Docs managing shared meeting minutes and updates. Weekly syncs via calls and email kept stakeholders engaged, while a WhatsApp group handled quick queries and day-to-day alignment.

Individual Reflections



Sinai Rhodes

We improved team communication and accountability through regular meetings. Despite challenges, effective planning and collaboration kept us on track, ensuring a well-refined final concept.



Freddie Nicholson

This project was challenging at times due to working in a different timeframe than usual. However, with guidance from our tutor and guest speakers, I now understand how to apply this in the workplace.



Emmanuel Irechukwu

I found the concept development phase to be highly enriching, as it enhanced my comprehension of the crucial elements that define our product. Additionally, it deepened my understanding of ADHD, providing me with valuable insights into the complexities associated with this condition.



Liberty Wright

I enjoyed getting to lead the reflection component of the project through user research. Balancing the project with other deadlines and extracurriculars has been a challenge, but super rewarding.



Kaitai Yang

This project deepened my understanding of haptic feedback and broadened my perspective on future design engineering, helping me think more strategically about long-term innovation.

References

Problem / Contextual Trends / Futures Wheel

1. University College London. Significant rise in ADHD diagnoses in the UK [Internet]. Available from: <https://www.ucl.ac.uk/news/2023/jul/significant-rise-adhd-diagnoses-uk>
2. PLoS One. Underdiagnosis of ADHD in adults [Internet]. Available from: <https://pmc.ncbi.nlm.nih.gov/articles/PMC4757442/pdf/pone.0148841.pdf>
3. King's College London. Neurofeedback may not be effective in reducing ADHD symptoms [Internet]. Available from: <https://www.kcl.ac.uk/news/neurofeedback-may-not-be-effective-in-reducing-adhd-symptoms>
4. National Foundation for Educational Research. Teacher labour market in England: Annual report 2024 [Internet]. Available from: <https://www.nfer.ac.uk/publications/teacher-labour-market-in-england-annual-report-2024/?web=1>
5. SEND Network. Autumn Budget 2024: Rachel Reeves boosts SEND funding by £1bn [Internet]. Available from: <https://send-network.co.uk/posts/autumn-budget-2024-rachel-reeves-boosts-send-funding-by-1bn>
6. APA PsycNet. Emotional dysregulation in ADHD: Implications for clinical practice [Internet]. Available from: <https://psycnet.apa.org/record/2021-53551-001>
7. University of Auckland. Bioengineers show that fidgeting might help concentration [Internet]. Available from: <https://www.auckland.ac.nz/en/news/2021/03/22/bioengineers-shows-that-fidgeting-might-help-concentration.html>
8. BBC News. ADHD in adults: Why are diagnoses rising? [Internet]. Available from: <https://www.bbc.co.uk/news/newsbeat-39855894>

Project Definition / Forecasting ADHD Span

1. The onset of attention-deficit hyperactivity disorder. Impuls. [Internet]. Available from: <https://www.sciencedirect.com/science/article/pii/S0197018615000212>
2. NHS. ADHD: Treatment and management [Internet]. Available from: <https://www.nhs.uk/conditions/attention-deficit-hyperactivity-disorder-adhd/treatment/>
3. Behavioral treatments for kids with ADHD. Child Mind Institute [Internet]. Available from: <https://childmind.org/article/behavioral-treatments-kids-adhd/#:~:text=Behavioral%20treatments%20for%20ADHD%20can,behavior%20problems%20do%20especially%20well.>
4. Reasonable adjustments for students with ADHD. Recite Me [Internet]. Available from: <https://reciteme.com/news/reasonable-adjustments-for-students-with-adhd/#:~:text=One%20of%20the%20most%20common,t%20reflect%20their%20true%20abilities.>

Pedagogy

1. ADHD Foundation. Teaching and managing students with ADHD [Internet]. Available from: <https://www.adhdfoundation.org.uk/wp-content/uploads/2022/03/Teaching-and-Managing-Students-with-ADHD.pdf>
2. ADDitude Magazine. Rejection sensitive dysphoria and ADHD [Internet]. Available from: <https://www.additudemag.com/rejection-sensitive-dysphoria-adhd-emotional-dysregulation/>

Concept Overview

1. Association of Paediatric Chartered Physiotherapists. Guide to use of electrical stimulation in paediatric neurodisability [Internet]. Available from: <https://apcp.csp.org.uk/content/guide-use-electrical-stimulation-paediatric-neurodisability>
2. Teach Me Anatomy. Muscles of the hand [Internet]. Available from: <https://teachmeanatomy.info/upper-limb/muscles/hand/>
3. Lab P. BackHandEMS: Wearable electrical stimulation for haptic feedback [Internet]. Available from: <https://lab.plopes.org/published/2021-CHI-BackHandEMS.pdf>

Biometrics

1. iMotions. Skin conductance response in biometrics research [Internet]. Available from: <https://imotions.com/blog/learning/best-practice/skin-conductance-response/>
2. IEEE Xplore. Advances in biometric monitoring [Internet]. Available from: <https://ieeexplore.ieee.org/document/8751953>
3. BBC. How to reset your brain with breathing [Internet]. Available from: <https://www.bbc.co.uk/programmes/articles/1mW6885X3N2gKnVjXT00KCj/how-to-reset-your-brain-with-your-breathing>
4. George Washington University. Mechanisms linking brain alertness and heart rate [Internet]. Available from: <https://smhs.gwu.edu/news/gw-researcher-discovers-mechanisms-link-brain-alertness-and-increased-heart-rate>
5. Analog Devices. Clinical-grade PPG in wearables [Internet]. Available from: <https://www.analog.com/en/resources/technical-articles/need-clinical-grade-ppg-from-your-wearable.html>

EMS & Ultrasound Haptics

1. IEEE Spectrum. SkinHaptics: Ultrasound haptic feedback inside the body [Internet]. Available from: <https://spectrum.ieee.org/skinhaptics-uses-ultrasound-to-generate-haptic-feedback-inside-your-body>

Battery / Power

1. Electro-Medical. EMS2000 Manual [Internet]. Available from: <https://electro-medical.com/images/EMS2000%20Manual.pdf>
2. MIT. Ultrasonic haptic feedback: Final project report [Internet]. Available from: https://web.mit.edu/6.101/www/s2017/projects/spohorec_Project_Final_Report.pdf
3. Census at School Canada. Wrist circumference data [Internet]. Available from: <https://censusatschool.ca/data-results/2006-07/wrist-circumference/>
4. Battery LiPo. Liter-703535 battery specifications [Internet]. Available from: <https://www.battery-lipo.com/index.php?c=category&id=31>
5. Statista. Global battery storage capacity projections [Internet]. Available from: <https://www.statista.com/statistics/1307203/world-battery-storage-electricity-generation-capacity/>
6. National Center for Biotechnology Information. Advances in wearable battery technology [Internet]. Available from: <https://pmc.ncbi.nlm.nih.gov/articles/PMC5683854/>

Economics

1. Custom Market Insights. Disabled and elderly assistive technology market report [Internet]. Available from: <https://www.custommarketinsights.com/report/disabled-and-elderly-assistive/>
2. Cambridge University Press. ADHD diagnoses and prescriptions in UK primary care [Internet]. Available from: <https://www.cambridge.org/core/journals/bjpsych-open/article/attentiondeficit-hyperactivity-disorder-diagnoses-and-prescriptions-in-uk-primary-care-20002018-populationbased-cohort-study/E0818CCAE895FF273C7448756CFB0066>
3. UK Government. New specialist places to be created in mainstream schools [Internet]. Available from: <https://www.gov.uk/government/news/new-specialist-places-to-be-created-in-mainstream-schools>
4. Printed Electronics Market By Technology (Flexography, Inkjet Printing, Screen Printing, Offset Printing, Others); By Material (Substrates, Inks) Read more at: <https://www.credenceresearch.com/report/printed-electronics-market>